

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO4: Analyze big data characteristics and challenges across domains including security and application aspects. (BI.4)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 100

95.1
20/8/2020

Code of Conduct

I Thamaraivelvi S (192511040) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thamaraivelvi S

Assessment Tasks

Class Test

1. Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.
2. Explain the 5Vs of Big Data with one practical example for any two Vs.
3. Differentiate structured, semi-structured, and unstructured data with examples.
4. Explain the role of distributed file systems in big data processing.
5. List any three major security concerns in big data platforms and briefly explain each.
6. What are data appliances and integration tools? Explain their role in big data workflows.
7. Discuss two major challenges organizations face while implementing big data analytics.
8. Write short notes on any one domain application of big data analytics.
9. How do business drivers influence the adoption of big data solutions?
10. Explain the importance of big data analytics in decision making.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful	Some relevant examples	Weak examples	No useful examples

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		examples			
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1) → Big Data refers to extremely large, complex, and rapidly generated datasets that cannot be efficiently stored, processed, or analyzed using traditional data processing systems.

→ Traditional systems are often insufficient because they have limitations in storage capacity, processing speed, scalability, and handling different data formats. Big Data requires distributed systems that can process huge amounts of data across multiple computers.

2) The 5Vs of Big Data are:

- i) Volume - Refers to the huge amount of data generated and stored.
- ii) velocity - Refers to the speed at which data is generated and processed.
- iii) variety - Refers to different types and formats of data.
- iv) veracity - Refers to the accuracy, reliability and quality data.
- v) Value - Refers to the useful information and benefits from data.

Examples:

- * volume: An e-commerce website stores millions of customer transactions every day.

* velocity: stock market systems receive and process thousands of transactions every second.

3)

Type	Description	Example
Structured Data	organized in a fixed format such as rows and columns.	SQL tables, bank records, employee details.
semi-structured Data	Does not follow a strict table structure but contains tags or metadata.	JSON, XML, emails
Unstructured Data	Has no predefined structure or format	Images, videos, audio, social media posts.

Example:

A customer database is structured, a JSON customer profile is semi-structured, and a customer review video is unstructured.

- 4) → A Distributed file system (DFS) stores large datasets across multiple computers or nodes in a network. Instead of storing all data on one machine, the data is divided into smaller blocks and distributed across different nodes.
- DFS provides scalability, fault tolerance, high availability, and faster data access. A common example is Hadoop Distributed file system (HDFS). It allows big data applications to store and process very large datasets efficiently.
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5) i) Data privacy:
Sensitive information such as customer and financial data must be protected from unauthorized access.

ii) unauthorized Access:

Attackers or unauthorized users may access important data. Authentication and Access-control mechanisms are required.

iii) Data Breaches:

Data may be stolen or exposed due to cyber attacks, weak security, or system vulnerabilities. Encryption and monitoring can help prevent breaches.

6) 8) Data Appliances :

→ Data Appliances are specialized hardware or software systems designed to store, manage, and process large amounts of data efficiently.

9) Data Integration tools :

→ Data integration tools help collect data from different sources and move or transform it into a common system for analysis.

They are important because big data usually comes from databases, applications, sensors, websites, and social media. Integration tools combine this data and make it available for analytics.

Example :

Apache Nifi, Talend, and Informatica can be used for data integration.

7) Data Quality :

Data collected from different sources may contain errors, duplicates, missing values, or inconsistent formats. Poor-quality data can produce inaccurate analytical results.

ii) Security and privacy:

organizations often handle sensitive customer and business information. protecting this data from unauthorized access and complying with privacy requirements can be ~~handling~~.

→ other challenges include are high implementation cost, lack of skilled professionals, scalability, and data management complexity.

8) Health Care:

→ Big Data Analysis is widely used in healthcare to analyse patient records, medical images, laboratory results, and treatment information. It helps doctors identify disease patterns, predict health risks, and improve treatment decisions.

→ Hospitals can also use analytics to manage resources, monitor patients and reduce operational costs, Thus big data helps improve healthcare quality, efficiency, and decision making.

- 9) → Business drivers are the factors that encourage organizations to adopt Big Data technologies. Companies use big data to reduce costs, improve customer satisfaction, increase revenue, identify new opportunities and gain competitive advantage.
- For example, an e-commerce company can analyze customer purchasing behaviour to provide personalized recommendations.
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- 10) → Big Data Analytics helps organizations make faster, more accurate, and data-driven decisions. It analyzes large datasets to identify patterns, trends, relationships, and future opportunities.
- Therefore, Big Data Analytics supports better planning, risk management, customer understanding, cost reduction, and business growth.
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